

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA0101
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.* (BTL 3)

Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I M. Veera Naga Sai (192425253) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Veera Naga Sai

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	Smart Healthcare Diagnosis	A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention. Tasks: (i) Identify facts, entities, predicates, variables, and constants. (ii) Represent the knowledge using Propositional Logic. (iii) Represent the knowledge using FOL with suitable quantifiers. (iv) Perform unification for two relevant predicates. (v) Apply resolution to prove whether Ravi has a viral infection. (vi) Show the complete inference sequence and final conclusion.
2	Student Academic Advising	A university AI system recommends academic support based on attendance and examination performance. Priya has attendance below 75% and has failed two courses. Students with low attendance require attendance counseling. Students who fail two or more courses require academic counseling. Students requiring both types of counseling need academic support. Tasks: (i) Identify the knowledge components. (ii) Convert the statements into Propositional Logic. (iii) Develop FOL representations using predicates and quantifiers. (iv) Identify suitable substitutions and perform unification. (v) Use resolution to prove whether Priya requires academic support. (vi) Explain the reasoning steps leading to the conclusion.
3	Intelligent Loan Approval	A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval. Tasks: (i) Identify entities, facts, predicates, and rules. (ii) Map the scenario into Propositional Logic. (iii) Represent the rules using FOL.

		(iv) Demonstrate unification and substitution. (v) Use resolution to prove or disprove LoanApproved(Arun). (vi) Provide the final inference with justification.
4	Vehicle Fault Diagnosis	An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service. Tasks: (i) Extract the facts and rules from the scenario. (ii) Represent them using Propositional Logic. (iii) Develop suitable FOL predicates and rules. (iv) Perform unification for the vehicle-related predicates. (v) Apply resolution to determine whether Car101 has a battery problem. (vi) Derive whether Car101 should be marked for service and explain the inference.
5	Cybersecurity Threat Detection	An organization uses an AI system to identify cyber threats. User101 has five failed login attempts and logged in from an unusual location. Multiple failed login attempts indicate a possible brute-force attack. A brute-force attack combined with an unusual login location indicates suspicious activity. Suspicious activity combined with unauthorized access to confidential files indicates a possible security breach. Tasks: (i) Identify the facts, entities, predicates, and rules. (ii) Map the knowledge into Propositional Logic. (iii) Represent the rules using FOL. (iv) Demonstrate unification and substitution. (v) Apply resolution to determine whether User101 is involved in a possible security breach. (vi) Present the complete reasoning chain and final inference.

LOGIC REASONING & MAPPING EXERCISE

SMART HEALTHCARE DIAGNOSIS

1. Problem Analysis and Knowledge Identification

A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. The system uses logical rules to determine the possible condition of the patient.

The given information contains the following facts:

1. Ravi has fever.
2. Ravi has cough.
3. Ravi has body pain.

The given rules are:

1. Patients with fever and cough may have flu.
2. Patients with flu and body pain may have viral infection.
3. Patients with viral infection require medical attention.

The main objective is to determine whether Ravi has a viral infection.

2. Logical Representation and Mapping

The knowledge can first be represented using propositional symbols.

Let:

F = Ravi has fever

C = Ravi has cough

B = Ravi has body pain

L = Ravi has flu

V = Ravi has viral infection

M = Ravi requires medical attention

Propositional Facts

F

C

B

Propositional Rules

Rule 1:

$(F \wedge C) \rightarrow L$

Rule 2:

$(L \wedge B) \rightarrow V$

Rule 3:

$$V \rightarrow M$$

Therefore, the complete propositional knowledge base is:

$$KB = \{F, C, B, (F \wedge C) \rightarrow L, (L \wedge B) \rightarrow V, V \rightarrow M\}$$

This representation shows that the initial symptoms can be used to derive further medical conditions.

3. Predicates, Variables, Constants and Quantifiers

First-Order Logic is used to represent the properties of patients and the general rules of the healthcare system.

Predicates

Fever(x) = x has fever

Cough(x) = x has cough

BodyPain(x) = x has body pain

Flu(x) = x has flu

ViralInfection(x) = x has viral infection

MedicalAttention(x) = x requires medical attention

Entity

The main entity in the problem is:

Ravi – the patient.

Constant

Ravi is a constant because it represents a specific patient.

Variable

x is a variable representing any patient.

Quantifier

The universal quantifier ($\forall$) is used because the rules are applicable to all patients.

FOL Facts

Fever(Ravi)

Cough(Ravi)

BodyPain(Ravi)

FOL Rules

Rule 1:

$$\forall x [(Fever(x) \wedge Cough(x)) \rightarrow Flu(x)]$$

Rule 2:

$$\forall x [(Flu(x) \wedge BodyPain(x)) \rightarrow ViralInfection(x)]$$

Rule 3:

$$\forall x [ViralInfection(x) \rightarrow MedicalAttention(x)]$$

4. Logical Connectives and Formula Construction

The following logical connectives are used in the knowledge representation.

$\wedge$ = AND

$\rightarrow$ = IMPLIES

$\neg$ = NOT

$\forall$ = FOR ALL

Formula 1

$\text{Fever}(x) \wedge \text{Cough}(x)$

This means that a patient has both fever and cough.

Formula 2

$(\text{Fever}(x) \wedge \text{Cough}(x)) \rightarrow \text{Flu}(x)$

This means that if a patient has fever and cough, the patient may have flu.

Formula 3

$(\text{Flu}(x) \wedge \text{BodyPain}(x)) \rightarrow \text{ViralInfection}(x)$

This means that if a patient has flu and body pain, the patient may have a viral infection.

Formula 4

$\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)$

This means that a patient with viral infection requires medical attention.

Therefore, the complete logical relationship is:

$\text{Fever} + \text{Cough} \rightarrow \text{Flu}$

$\text{Flu} + \text{Body Pain} \rightarrow \text{Viral Infection}$

$\text{Viral Infection} \rightarrow \text{Medical Attention}$

5. Inference and Logical Reasoning

Inference is the process of deriving new information from existing facts and rules.

Step 1: Initial Facts

The system initially knows:

$\text{Fever}(\text{Ravi})$

$\text{Cough}(\text{Ravi})$

$\text{BodyPain}(\text{Ravi})$

Step 2: Apply Rule 1

Rule 1:

$\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$

Since Ravi has fever and cough:

$\text{Fever}(\text{Ravi}) \wedge \text{Cough}(\text{Ravi})$

Therefore:

Flu(Ravi)

Step 3: Apply Rule 2

Rule 2:

$\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$

The system now knows:

Flu(Ravi)

BodyPain(Ravi)

Therefore:

ViralInfection(Ravi)

Step 4: Apply Rule 3

Rule 3:

$\text{ViralInfection}(x) \rightarrow \text{MedicalAttention}(x)$

Since:

ViralInfection(Ravi)

Therefore:

MedicalAttention(Ravi)

Complete Inference Sequence

Fever(Ravi) + Cough(Ravi)

↓

Flu(Ravi)

↓

Flu(Ravi) + BodyPain(Ravi)

↓

ViralInfection(Ravi)

↓

MedicalAttention(Ravi)

Thus, the system concludes that Ravi may have a viral infection and requires medical attention.

6. Unification and Substitution

Unification is the process of finding suitable substitutions for variables so that two predicates become identical.

Example 1

Consider the following predicates:

Fever(x)

Fever(Ravi)

To make the two predicates identical, substitute:

$x = \text{Ravi}$

After substitution:

$\text{Fever}(\text{Ravi}) = \text{Fever}(\text{Ravi})$

Therefore, the predicates are successfully unified.

Most General Unifier:

$\{x/\text{Ravi}\}$

Example 2

Consider:

$\text{BodyPain}(x)$

$\text{BodyPain}(\text{Ravi})$

Substitute:

$x = \text{Ravi}$

After substitution:

$\text{BodyPain}(\text{Ravi}) = \text{BodyPain}(\text{Ravi})$

Therefore, the predicates are successfully unified.

Most General Unifier:

$\{x/\text{Ravi}\}$

Thus, unification allows the general rules to be applied to the specific patient Ravi.

7. Resolution and Proof Construction

The selected query is:

Does Ravi have a viral infection?

Query:

$\text{ViralInfection}(\text{Ravi})$

To prove the query using resolution, the negation of the query is added to the knowledge base.

Negated Query:

$\neg \text{ViralInfection}(\text{Ravi})$

Convert Rule 1 into Clause Form

Original rule:

$\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$

Clause form:

$\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$

Convert Rule 2 into Clause Form

Original rule:

$\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$

Clause form:

$\neg \text{Flu}(x) \vee \neg \text{BodyPain}(x) \vee \text{ViralInfection}(x)$

Given Facts

Fever(Ravi)

Cough(Ravi)

BodyPain(Ravi)

Resolution Step 1

Using:

Fever(Ravi)

Cough(Ravi)

and:

$\neg \text{Fever}(\text{Ravi}) \vee \neg \text{Cough}(\text{Ravi}) \vee \text{Flu}(\text{Ravi})$

we obtain:

Flu(Ravi)

Resolution Step 2

Using:

Flu(Ravi)

BodyPain(Ravi)

and:

$\neg \text{Flu}(\text{Ravi}) \vee \neg \text{BodyPain}(\text{Ravi}) \vee \text{ViralInfection}(\text{Ravi})$

we obtain:

ViralInfection(Ravi)

Resolution Step 3

We now have:

ViralInfection(Ravi)

and:

$\neg \text{ViralInfection}(\text{Ravi})$

Resolving these two statements produces:

□

The empty clause indicates a contradiction.

Therefore, the negation of the query is false.

Hence, the original query is true:

ViralInfection(Ravi)

Therefore, Ravi is inferred to have a viral infection according to the given knowledge base.

8. Mapping Diagram / Clarity and Presentation

The complete mapping of the knowledge can be represented as:

Ravi has Fever

+

Ravi has Cough

↓

Ravi may have Flu

↓

Ravi has Body Pain

↓

Ravi may have Viral Infection

↓

Ravi requires Medical Attention

Complete Inference Flow

Fever(Ravi) + Cough(Ravi)

↓

Flu(Ravi)

↓

Flu(Ravi) + BodyPain(Ravi)

↓

ViralInfection(Ravi)

↓

MedicalAttention(Ravi)

Final Conclusion

The AI system starts with the known symptoms of Ravi: fever, cough, and body pain. From fever and cough, the system infers that Ravi may have flu. From flu and body pain, it further infers that Ravi may have a viral infection. Finally, since patients with viral infection require medical attention, the system concludes that Ravi requires medical attention.

Therefore, the final inference is:

Ravi has Fever + Cough + Body Pain

→ Ravi may have Flu

→ Ravi may have Viral Infection

→ Ravi requires Medical Attention.

EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.

		construct valid logical formulas.			
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram & Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.